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Quality Improvement

Patient safety: a focused associate-degree nursing learning module

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ABSTRACT

Background: The number of medical errors in the United States is estimated to be up to 250,000 per year. Nurses are critical to patient safety; however, few nursing programs include curricula focused on system-level elements that foster patient safety or experiential training on graded assertiveness communication techniques.
Aim: This study examined the effectiveness of a safety-focused educational intervention for 22 students in an associate-degree nursing (ADN) program at a rural technical college in the Southeastern United States.
Methods: The investigator used videos, lectures, discussions, and role-play to teach students about system-level elements that foster patient safety and a graded assertiveness communication technique. To evaluate effectiveness, a survey measured students' self-reported knowledge of patient safety concepts and levels of confidence in voicing safety concerns.
Results: Statistical analysis of aggregated pre- and post- intervention survey results demonstrated significant gains in both self-reported knowledge of patient safety concepts and levels of confidence in voicing safety concerns.
Conclusions: Findings suggest that a brief but focused educational module such as the one described may be a suitable and effective approach to improving ADN students' knowledge and confidence regarding patient safety.

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Due to their central function in the provision of patient care, registered nurses (RN) play a critical role in patient safety. Thus, it is vital that all nurses, including associate-degree nurses (ADN), have a strong understanding of patient safety concepts upon graduation and feel confident in speaking up should they identify safety concerns.

Nearly 36% of nurses in the United States enter into the profession with an ADN (Smiley et al., 2023). However, a number of studies have demonstrated that, in hospital settings, the risk of patient mortality is reduced when the percent of BSN-prepared nurses is greater (Aiken et al., 2003, 2011, 2014; Cho et al., 2015; Dierkes et al., 2021; Estabrooks et al., 2005; White et al., 2018). Similar reductions were seen in failure to rescue events (Aiken et al., 2003; White et al., 2018).

Although the root cause(s) of these correlations are nuanced and likely very complex, Anbari et al. (2017) found that nurses who went through a traditional BSN program viewed patient safety from a broader and more complex systems-level perspective than those who went through an ADN-to-BSN program. These findings may suggest that safety education is presented differently in the two degree programs. Despite these disparities, graduates of both degree programs enter into nursing with the same license, scope of practice, and responsibilities.

At the rural Southern technical college that served as the setting for this study, the ADN curriculum does not include a learning module specifically focused on patient safety and/or communication about safety concerns; instead, safety and communication concepts are embedded throughout the program. This project aimed to explore the suitability and effectiveness of a safety-focused learning strategy for a rural technical college ADN program by answering the following: Among final-semester associate-degree nursing students, does a brief but focused educational intervention improve knowledge of patient safety concepts and increase confidence to voice safety concerns?

This project had two objectives. The first was to improve the students' knowledge about patient safety; a second and equally important objective was to improve the students' confidence to speak up about patient safety concerns. Increased knowledge and confidence in this area has the potential to improve safety for all of the students' future patients. Further, if participants become nursing leaders or educators in the future, the intervention could ultimately have a positive impact on safety culture in their organizations and on future generations of nurses.

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Background

Medical Errors

In 1999, the Institute of Medicine (IOM) published *To err is human: Building a safer health system*. This landmark publication revealed the shocking statistics regarding medical errors that occur in the U.S. healthcare system, prompting a national focus on patient safety. At the time, investigators reported that as many as 98,000 people died each year due to preventable medical errors (IOM, 2000). However, a more recent study estimated that the number is closer to 250,000, making it the third leading cause of death (Makary & Daniel, 2016). Nurses play a key role in advocating for patients; therefore, it is critical that all have a strong understanding of both individual and system-level safety concepts and feel confident to voice concerns.

Current Integration of Safety Education

In response to the 1999 IOM report, the Quality and Safety Education in Nursing (QSEN) competencies were developed to ensure that nurses have the knowledge, skills, and attitudes (KSA) necessary to promote patient safety (Altmiller & Hopkins-Pepe, 2019). These competencies are required in nursing programs accredited by the American Association of Colleges of Nursing (AACN) and the Accrediting Commission for Education in Nursing (ACEN). However, a recent systematic review revealed few evidence-based educational interventions focused on patient safety (Lee et al., 2022). Moreover, studies show patient safety has been integrated inconsistently into program curricula across the United States (Altmiller & Armstrong, 2017; Cengiz & Yoder, 2020). Further, although the broader concepts of a *systems* approach support the culture essential to the consistent provision of high quality, safe, and efficient patient care (Leape, 2021), most nursing curricula focus on *individual* aspects of safety, rather than exploring the effects of systems on patient safety (Lee et al., 2022).

Confidence in Speaking Up for Patient Safety

Breakdowns in communication are a leading factor in patient harm events (The Joint Commission, 2023). Accordingly, communication should be incorporated into lessons focused on patient safety. Individual, program, and facility-level factors can influence the willingness to voice concerns (Palese et al., 2018), as can social influences such as the desire to fit in (Fisher & Kiernan, 2019). Actual or perceived authority gradients between and within disciplines can also hinder communication due to factors such as intimidation, worries of sounding inexperienced, or fear of retribution (Hubbard & Chicca, 2022).

Educational Intervention Studies

Although it is common for many nursing programs to thread patient safety topics and communication skills throughout the curriculum (Lee et al., 2022), a number of small cross-sectional studies have demonstrated measurable improvement in student KSAs after various focused educational interventions. For example, Kim et al. (2019) demonstrated an increase in student safety competency using a flipped classroom technique; however, the intervention required a total of 28 hours of classroom time, which is unlikely to be feasible in most ADN programs. Jamshidi et al. (2021) used a problem-based learning method, which also resulted in higher student levels of knowledge, attitudes, and perceptions on patient safety compared to a traditional didactic approach. Noh and Kim (2021) found that teaching students a structured communication technique (Situation, Background, Assessment, Recommendation) along with assertiveness training improved communication skills and clinical competence, and lowered students' stress levels during clinical training.

Finally, Hanson et al. (2020) had students watch a video reenactment of an actual event in which a series of failed communications resulted in the preventable death of a 37-year-old mother of two. Following the video, researchers facilitated reflective discussion and role-play using structured communication techniques. In this study, researchers determined that training nursing students on the concepts of "speaking up for safety" increased their confidence and feelings of empowerment.

Notably, all studies were conducted in a university setting with BSN students. A literature review revealed no relevant research on the effectiveness of similar educational interventions conducted within ADN programs.

Methods

Context

The setting for this quality improvement project was a small public technical college located in a mountain community of North Georgia that offers literacy, diploma, certificate, and associate-degree programs, including nursing. The target population was one cohort ($N = 23$) of final-semester ADN nursing students preparing for entry into professional practice. All but one of the students completed both the pre- and postsurveys, yielding a total of 22 participants.

Permissions and Ethical Considerations

The project proposal was reviewed by the principal investigator's institutional review board prior to implementation, and was determined to be exempt. All procedures were performed in compliance with institutional guidelines, and the methods for administering the survey were designed to safeguard the anonymity of participants and minimize potential breaches of confidentiality. Permission to conduct the project was obtained from the ADN Program Director of the technical college where the intervention was conducted.

Intervention

Gagné's Nine Events of Instruction was utilized to provide a definitive structure for the development of a succinct yet highly effective curriculum. The program was modeled after the previously described communication training course employed by Hanson et al. (2020).

The educational intervention consisted of 1- to 2-hour weekly sessions over 4 weeks, integrated into regularly scheduled class time. During the first session, students completed the preintervention survey to assess their baseline knowledge and comfort level on the subject. Students were then shown three brief videos. The first, *Speaking Up Stories #1: Janice Palaganas' Burrito* (Center For Medical Simulation, 2019), is a 2-minute long, somewhat light-hearted reminder of how difficult it can be for *anyone* to speak up at times. The second, *Patient Story—Pressure Injury* (Clinical Excellence Commission, 2017) is narrated by two nurses describing the devastating harm their mother experienced as a result of missed nursing care. Finally, students watched 6 minutes of *QSEN 2017: 12—Patient Safety Speaker* (QSEN Institute, 2017, 24.40–30.48), in which patient safety expert Dr. Carol Durham presents the case of Lewis Blackman, a 15-year-old boy who died due to medical error. Dr. Durham also addresses the CUSS (concerned, uncomfortable, safety, stop) graded assertiveness technique and the Swiss-cheese model of medical errors. After each video, the facilitator reviewed important points and encouraged discussion among the students.

The next session started with a presentation of the program objectives, followed by a PowerPoint-supported lecture outlining the

elements of patient safety, with a focus on a systems perspective rather than individual caregiver responsibilities. Topics included the prevalence of medical errors, cultural and systems elements known to reduce errors, the impact of missed care, the Swiss-cheese model of safety events, and the value of reporting near-miss incidents. For interest and relevance, lectures were supplemented with real-life scenarios.

The third session focused on the CUSS model of graded assertiveness communication. The lesson utilized American Hospital Association (n.d.) materials, including a brief video explaining the technique as well as one providing an interactive demonstration of the communication tool in use. Following this, students were given scenarios and guided in role-play use of the technique. The previously described social and hierarchal challenges that often exist in the clinical setting were embedded in the scenarios. During the fourth and final session, student knowledge and confidence were reassessed using the same survey administered initially.

Measures

The impact of the educational intervention was assessed by comparing the results of the 18-question pre- and postintervention survey. All questions utilized a 5-point Likert scale. The survey included 8 questions that measured students' perceived level of *knowledge* about the patient safety concepts covered in the program and 10 questions that assessed their level of *confidence* to voice safety concerns. Responses within the knowledge and confidence sections of the survey were aggregated separately.

Within the knowledge section of the survey, the first 5 questions directly aligned with the educational objectives of the program. Questions 6 to 8 were adopted with permission from the patient safety competency self-evaluation (PSCSE) tool for nursing students developed by Lee et al. (2014). All of the remaining questions comprised the confidence section of the survey and were based on a questionnaire developed by Richard et al. (2017) to measure the "speak up" climate of healthcare organizations. The questions were modified slightly to be applicable to nursing students and to improve readability. Permission to modify and utilize the questions was obtained from the author of the tool. Four of these questions were written using negatively structured statements. Richard et al. (2017) utilized both positive and negative question formats to measure two separate behaviors: respondents' willingness to speak up and their tendency toward silence. The use of negatively worded questions can also serve to minimize acquiescence bias. To accurately reflect the intent of the question, scores for the negatively structured questions were tabulated in reverse.

Because the survey created for this project utilized select individual components of the tools created by Richard et al. (2017) and Lee et al. (2014), rather than one in its entirety, construct validity and/or reliability of the tool is unknown. The survey was, however, designed to carefully mirror the program content. To provide some level of face validity, the tool was also subjectively evaluated by three doctoral-prepared healthcare educators and two associated degree nurses for both clarity and content alignment.

Results

The revised standards for quality improvement reporting excellence (SQUIRE 2.0) were used as a framework for reporting. A comparison of pre- and postintervention average student scores demonstrated improvement in all 18 elements of the survey.

Knowledge

Students showed the most significant improvement in the knowledge section of the survey (Fig. 1). The mean initial scores on 3 of the elements were at or near "4," indicating concepts that were likely already embedded in the program; these included the consequences of missed nursing care, the importance of reporting near-miss events, and the definition of the term "medical error."

However, the knowledge section of the pretest also revealed considerable opportunities for learning, with average responses at or lower than mid-point (3) on half the questions. These lower-scoring questions included those centered on the influence of authority gradients on patient safety, system-level factors that create a culture of safety, the Swiss-cheese model of healthcare errors, and the prevalence of medical errors in the United States. The lower scores may indicate these were novel concepts to some or most of the students.

Upon completion of the program, responses showed favorable gains on every element in the knowledge section; postintervention

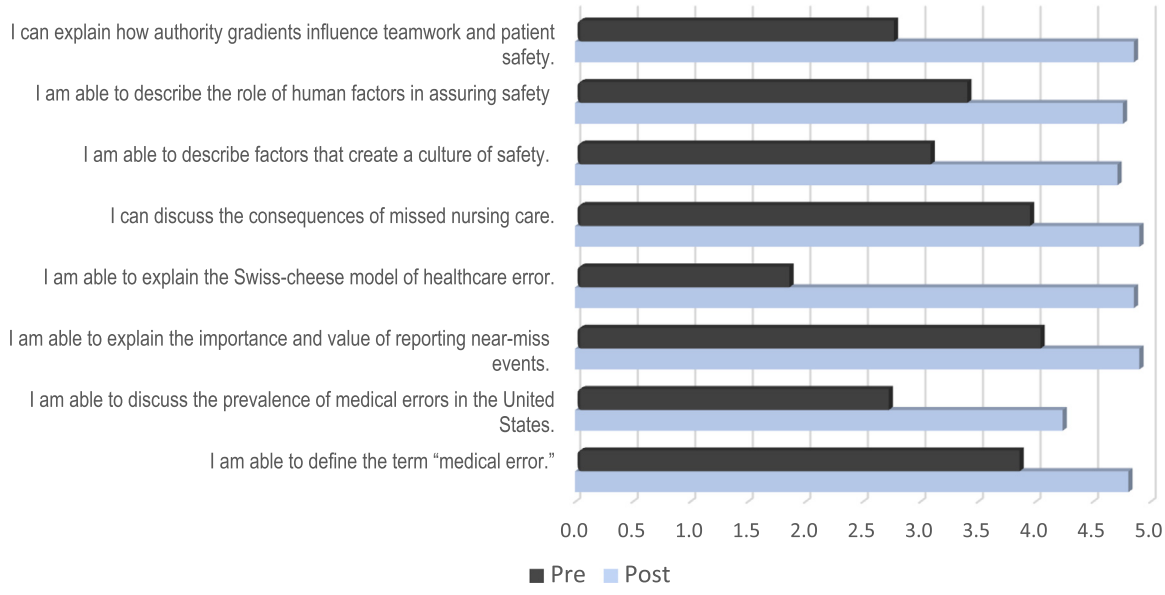


Fig. 1. Average self-reported knowledge of patient safety concepts ($n = 22$). *Note:* A comparison of pre- and postintervention self-reported knowledge of patient safety concepts using a 1 to 5 Likert scale (1 = not very knowledgeable to 5 = very knowledgeable).

Table 1
Analysis of unpaired *t* test.

Category	Presurvey		Postsurvey		*t*(df)	*p*
	M	SD	M	SD		
Knowledge	25.73	4.90	37.68	2.70	10.02 (42)	<0.001
Confidence	40.64	3.72	47.09	3.07	6.27 (42)	<0.001

Note: Results of an independent samples *t* test using aggregated results of pre- and postsurvey.

averages were at or above 4.24 for every question. Moreover, out of the possible 40 points in this section, the mean total score increased from 25.73 (SD = 4.90) preintervention to 37.68 (SD = 2.70) postintervention. Results of an unpaired *t* test (Table 1) demonstrated this improvement to be significant ($p < 0.001$).

Confidence

Impressively, the students' self-reported confidence level was high at the start of the program, with total scores averaging 40.64 (SD = 3.72) out of a possible 50 points. Nonetheless, the intervention had a positive impact in this area as well, with improvements observed on every question across the board, as shown in Fig. 2. Postintervention scores increased to 47.09 (SD = 3.07); an unpaired *t* test (Table 1) indicated the improvement in this section was also significant ($p < 0.001$).

The biggest improvement in self-reported confidence regarded the participants' willingness to remind a doctor about hand hygiene. Initially, this was the item about which the students reported having the lowest confidence (M = 3.05); however, scores on this element increased significantly postintervention (M = 4.43). An increase was anticipated as the educational sessions and discussions stressed that a nurse speaking up for patient safety is not only accepted, but expected, practice. Other notable increases showed in readiness to report a near-miss event and speaking directly to a colleague who failed to follow safety protocols; both means increased from 3.73 to 4.57.

Discussion

To effectively advocate for patients, all registered nurses, regardless of their level of education or experience, should be well-versed in matters of patient safety and have, at minimum, a basic level of comfort in voicing identified concerns. The positive results of this educational intervention in an ADN setting are similar to previous studies involving BSN students, including those conducted by Hanson et al. (2020), Jamshidi et al. (2021), Kim et al. (2019), and Noh and Kim (2021). The findings suggest that a focused educational intervention, requiring minimal time and no additional costs or resources, can improve the knowledge of patient safety concepts and the confidence to voice safety concerns among final-semester ADN students.

Strengths

One strength of the project lies in the fact that all participants were final-semester nursing students and therefore likely had sufficient experience in healthcare systems to understand some of the powerful human and system-level dynamics that can ultimately impact patient safety. The timing of the intervention was also intended to strengthen student knowledge and confidence prior to entering into professional practice. The timing of this

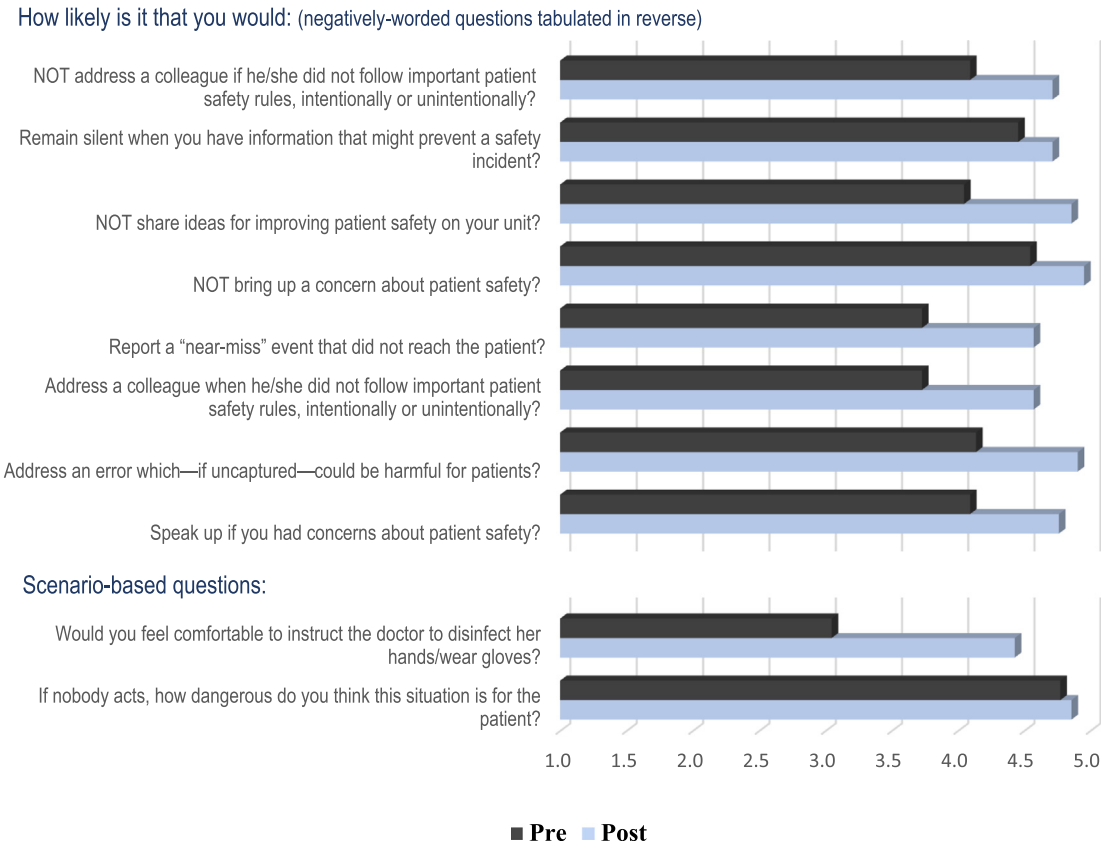


Fig. 2. Average self-reported confidence to speak up ($n = 22$). *Note:* A comparison of pre- and postintervention self-reported confidence to speak up for patient safety using a 1 to 5 Likert scale (1 = not likely to 5 = likely).

project differed from the one on which it was modeled by Hanson et al. (2020), in which researchers conducted the intervention *prior* to students' clinical placement, aiming to enhance student confidence and empowerment during clinical practicums. The approach by Hanson et al. also allowed students the opportunity to practice the skills learned during their clinical experiences. Arguably, both approaches have merits and future researchers may want to compare long-term outcomes using both approaches and/or a combination of the two (initial instruction early in the program with a reinforcement of the skills and concepts during the final semester).

Limitations

This project does have limitations that must be considered. First, the small sample size diminishes the strength of the findings. The geographically isolated sample also limits the generalizability of the findings, as such a limited pool of participants may not be a true representation of final-semester ADN students across the United States. Another limitation is that the project design did not have a mechanism to assess the possible impact of confounding variables such as the age, gender, ethnicity, geographical location, socioeconomic status, and previous experience of the participants. Finally, the pre- and postintervention survey relied solely on students' *perceived* levels of knowledge and confidence. Because such a measure is subjective, one cannot be confident as to the internal validity of the findings.

Implications for Nurse Educators

Despite these limitations, the present study provides a model upon which other ADN program leaders could implement similar safety-focused learning modules. Without doubt, patient safety should be embedded in all aspects of nursing education from the outset. However, the exposure that students have had to the healthcare setting by final semester may provide the perspective necessary to relate to the higher-level concepts of systems thinking. The module described in this paper was designed to effectively introduce a number of key concepts in an efficient manner that could be incorporated into an ADN curriculum. Additionally, the videos utilized in this program are easily accessed and can effectively supplement the content by providing a springboard for rich discussion.

Future Directions

It is hoped that the results of this study will prompt further research on this critical topic. Conducting a formal study on a larger scale with an increased pool of participants within a variety of ADN program settings would increase the reliability and generalizability of the findings. Additionally, conducting multiple intervals of measurement (e.g., upon completion of initial training, and after 3 and 12 months after job attainment) would help discern if students' perceived knowledge and level of comfort in speaking up learned from the intervention increases with experience or diminishes once a nurse is assimilated into an organization's culture. Future research could also strive to determine in which semester(s) such a program would provide the most impact over the long-term. Developing effective methods to deliver education focusing on patient safety in an ADN nursing curriculum is an important step in the long and complex journey to reduce medical errors and improve patient safety.

Author's Contributions

Maria Cuccinello: Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Data Curations, Writing (original draft), Writing (review & editing), Visualizations, Project Administrations. Lisa Fee: Methodology, Validation, Formal Analysis, Writing (review & editing), Supervision.

Declaration of Competing Interests

None to report.

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